Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 (emissions); atmospheric CO_2: Feb 2026

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

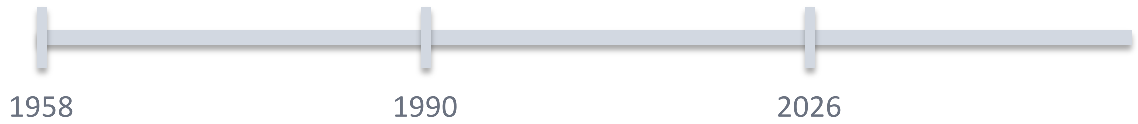
429 ppm

Mauna Loa (NOAA), February 2026

~280 ppm

Approx. pre-industrial baseline

Today’s concentration is >50% higher than pre-industrial levels (≈1.5×).
The recent rise is exceptionally rapid compared with natural post–ice-age changes.
Multiple measurement systems confirm acceleration, primarily driven by coal, oil, and gas combustion.



Acceleration since modern monitoring began (Mauna Loa record starts 1958)

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

~6 Bt

Global CO_2 emissions in 1950

>20 Bt

By 1990 (nearly quadrupled)

>35 Bt

Today (2024): fossil fuels + industry; not yet peaked

Visual: Global emissions line chart (1750–2024)

Show slow growth pre-1950, rapid post-1950 acceleration, and recent slowing without a peak.
Highlight inflection points: 1950 (~6 Bt), 1990 (>20 Bt), 2024 (>35 Bt).
Reference: figure_2.jpg (as specified).

Interpretation: Growth has slowed, but total annual emissions remain above 35 billion tonnes and have not yet peaked.

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

China: ~12.5 Bt/year (~27% of global); slight increase.
United States: ~5.0 Bt/year (~14%); declining from ~6 Bt peak around 2005.
India: ~3.0 Bt/year (~8%); rising.
EU-27: ~2.7 Bt/year (~7%); declining.
Context: China’s steep rise is concentrated post-2000.

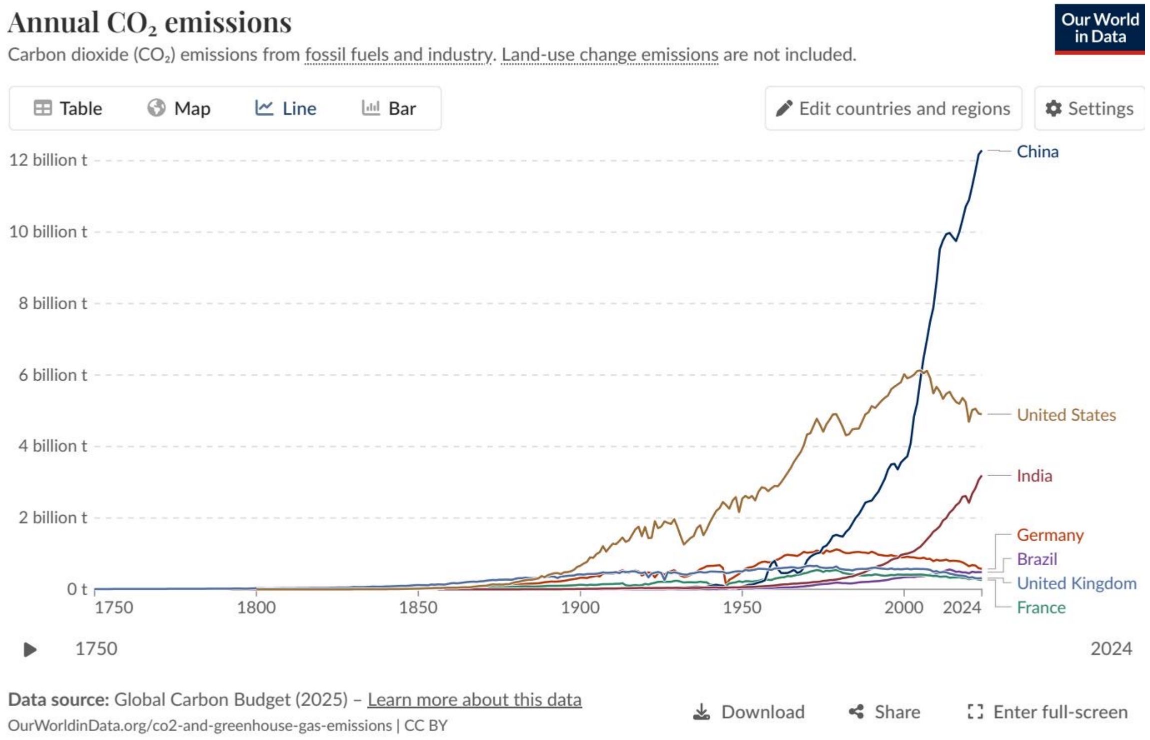


Figure reference: figure_3.jpg (Annual CO_2 emissions by country)

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

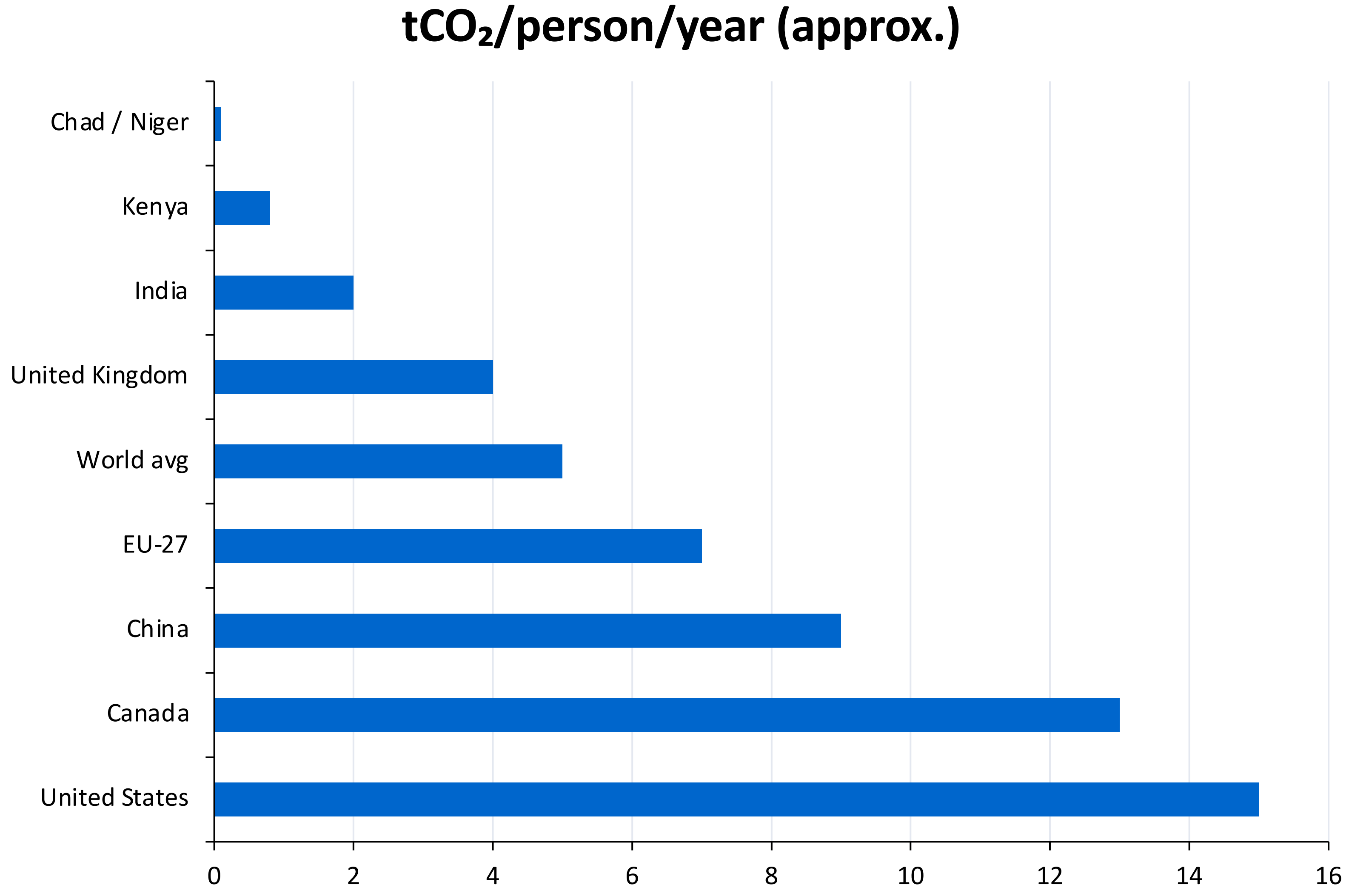
1900: Europe + United States accounted for >90% of global emissions.
1950: Europe + United States still >85%.
2024: Asia produces ~50% of emissions; US + Europe combined are <⅓.
Africa and South America each contribute ~3–4% (similar to international aviation + shipping combined).

Visual: Regional emissions shares (1900 vs 2024)

Use either:
• Stacked area chart: regional shares from 1900→2024; or
• Side-by-side 100% bars (1900 vs 2024) highlighting Asia's rise.
Key callouts: Europe+US collapse from >90% to <33%; Asia ~50%; Africa & S. America ~3–4% each.

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



Large, affluent economies can emit 2–3× the world average per person.
The per capita gap between high and low emitters is ~150-fold.
Framing matters for negotiations: totals drive warming, but per-person exposure highlights equity.

Figure reference: figure_1.jpg (per capita trend over time)

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

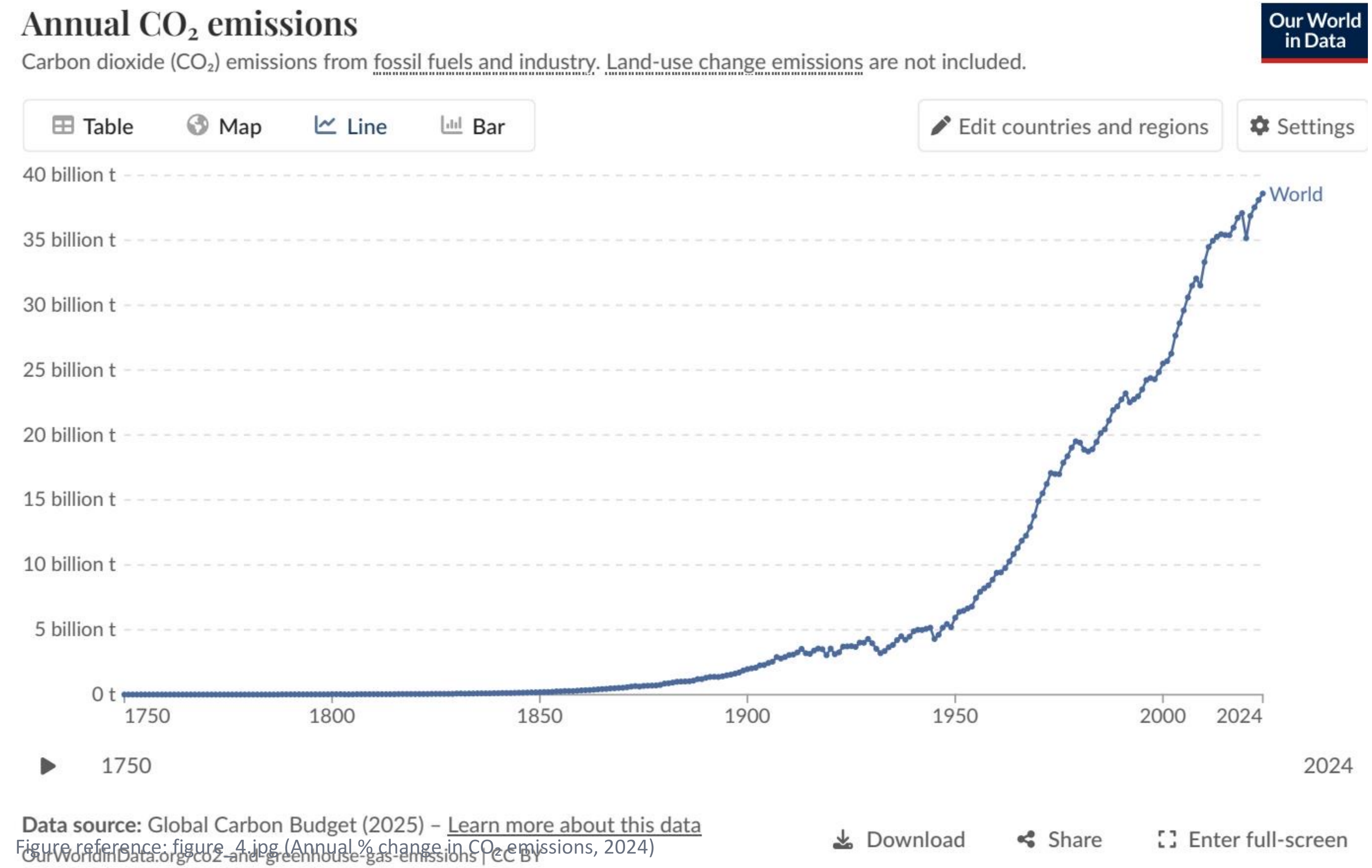


Figure reference: figure_4.jpg (Annual % change in CO_2 emissions, 2024)

US and much of Europe show declines (roughly −1% to −5%).

Russia, parts of Southeast Asia, and several African nations show increases (+2% to +10% or more).

South America is mixed: some countries show very large percentage increases (>10%), others decline.

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have materially different per capita CO_2 based on power and heat supply.
France and the UK: lower per capita CO_2, linked to higher nuclear and renewable shares.
Germany and the Netherlands: higher fossil fuel reliance contributes to higher per capita CO_2.

Negotiation relevance: energy system choices (power, heat, industry) can decouple prosperity from per-capita emissions.

Country group	Dominant low-carbon sources	Higher-carbon dependence	Per capita CO_2
France + UK	Higher nuclear and renewables	Lower fossil share (relative)	~4 t/person
Germany + Netherlands	Renewables present	Higher fossil fuel share	Higher than France/UK (qual.)

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

Global CO_2 emissions exceed 35 Bt/year and have not yet peaked, despite slower growth.
China emits ~12.5 Bt/year (~27%) — more than double the United States (~5 Bt).
Regional center of gravity shifted: Europe+US fell from >90% (1900) to <⅓ today; Asia ~50%.
Per-capita inequality remains extreme: ~150× gap between high and low emitters.
Policy and energy mix choices meaningfully shape per-capita outcomes among similarly prosperous countries.

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA/NOAA Mauna Loa